

AI Financial Analysis

Business Objective

Enable the Loan Origination System (LOS) to analyze the applicant's financial statements using AI, calculate key financial ratios, assess repayment capacity, and generate financial insights to support credit appraisal.

Scope

In Scope

- Financial statement upload
- Balance Sheet Analysis
- Profit & Loss Analysis
- Cash Flow Analysis
- Financial Ratio Calculation
- Working Capital Assessment
- Debt Service Capacity Analysis
- AI-generated Financial Summary

Out of Scope

- Credit Approval
 - CAM Generation
 - Loan Sanction
 - Disbursement
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Actors

- MSME Customer
 - AI Financial Analysis Engine
 - Loan Origination System
 - Credit Officer
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Business Rules

- Financial statements must belong to the applicant.
- Audited financial statements shall be preferred where available.
- AI shall calculate financial ratios based on uploaded financial data.

- AI recommendations are advisory and require Credit Officer review.
- Financial analysis results shall be stored against the Application Reference Number.
- All analysis activities shall be recorded in the audit trail.



Could not identify user

US-701 Retrieve Financial Statements

As the Loan Origination System,

I want to retrieve the customer's financial statements,

So that financial analysis can begin.

US-702 Analyze Financial Statements

As the AI Financial Analysis Engine,

I want to analyze the Balance Sheet, Profit & Loss Statement, and Cash Flow Statement,

So that the customer's financial health can be assessed.

US-703 Calculate Financial Ratios

As the AI Financial Analysis Engine,

I want to calculate key financial ratios,

So that repayment capacity and business performance can be evaluated.

US-704 Generate Financial Health Summary

As the AI Financial Analysis Engine,

I want to generate a financial health summary,

So that the Credit Officer receives actionable financial insights.

US-705 Generate AI Confidence Score

As the AI Financial Analysis Engine,

I want to generate a confidence score for the financial analysis,

So that the system can determine whether the AI analysis is reliable enough to continue automatically.

US-705 AI Explainability Report

As a Credit Officer,

I want the AI to explain the reasons behind its financial assessment,

So that I can understand and validate the recommendationUS-706 Route Low Confidence Applications for Manual Review

US-706 As the Loan Origination System,

I want to flag applications with a low AI confidence score for manual financial review,

So that potentially inaccurate AI recommendations do not influence lending decisions.

US-707 Evaluate AI Governance

As the AI Governance Engine,

I want to evaluate confidence, explainability, data quality, bias, and policy compliance,

So that only trustworthy AI outputs are used during credit appraisal.US-707 Proceed to Collateral Assessment

US-708 Detect Hallucination or Unsupported Conclusions

As the AI Governance Engine,
I want to identify unsupported AI-generated statements,
So that inaccurate financial recommendations are prevented.

US-709 Human Review Decision

As the Loan Origination System,
I want to automatically route loan applications for manual review whenever the AI Governance Evaluation identifies governance concerns such as low confidence, bias, missing financial data, hallucination, or insufficient explainability,
So that AI-generated financial insights are validated by a Credit Officer before they influence any credit appraisal decision.

US-710 Proceed to Collateral Statement

As the Loan Origination System,
I want to initiate collateral assessment after successful financial analysis,
So that the credit appraisal process can continue.

Acceptance Scenarios

Scenario 1: Retrieve Financial Statements

Given the customer has successfully completed regulatory verification,
When the financial analysis process begins,
Then the system shall retrieve the customer's uploaded financial statements.

Scenario 2: Successful Financial Analysis

Given valid financial statements are available,
When the AI analyzes the financial data,
Then the system shall calculate financial ratios and generate a Financial Health Summary.

Scenario 3: High AI Confidence

Given the AI has completed financial analysis,
When the AI Confidence Score is equal to or greater than the configured threshold,
Then

- the Financial Health Summary shall be stored,
 - the application shall automatically proceed to Collateral Assessment.
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Scenario 4: Low AI Confidence

Given the AI has completed financial analysis,

When the AI Confidence Score is below the configured threshold,

Then

- the application shall be flagged for manual financial review,
 - the Credit Officer shall be notified,
 - the workflow shall pause until the manual review is completed.
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Scenario 5: Missing Financial Statements

Given financial statements are unavailable,

When financial analysis is initiated,

Then

- the system shall request the customer to upload the required financial statements,
 - financial analysis shall not proceed until the required documents are available.
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Scenario 6: Bias Detected

Given the AI Governance Evaluation has been completed,

When bias indicators exceed the bank's configured threshold,

Then

- the application shall be flagged,
- the bias assessment report shall be attached,
- the Credit Officer shall review the recommendation before proceeding.

Scenario 4: Hallucination or Unsupported Conclusions Detected

Given the AI Governance Evaluation has completed,

When unsupported financial conclusions or hallucinations are detected,

Then

the AI recommendation shall be rejected,

the application shall be routed for manual review,

the governance issue shall be recorded in the audit log.

Scenario 5: Explainability Unavailable

Given the AI has generated a financial assessment,

When the AI cannot provide sufficient explainability for its recommendation,

Then

the recommendation shall not be considered for automated workflow progression,

the application shall be routed to the Credit Officer for manual assessment,

the explainability issue shall be recorded for governance and audit purposes.

Scenario 6: Governance Evaluation Passed

Given the AI Governance Evaluation has been completed,

When

the AI Confidence Score meets the configured threshold,

no material bias is detected,

required financial data is complete,

no hallucinations are detected, and

explainability requirements are satisfied,

Then

the AI recommendation shall be marked as governance approved,

the Financial Health Summary shall be stored,

the application shall automatically proceed to the next stage of the Loan Origination workflow.

Non-Functional Requirements

- Financial analysis shall complete within the bank's configured response time.
- AI calculations shall be accurate, auditable, and repeatable.
- Financial documents shall remain encrypted during storage and processing.
- All AI-generated outputs shall be version-controlled.
- Confidence score generation shall use approved bank models and configuration parameters.
- All analysis results shall be logged for audit and compliance purposes.

Dependencies

- Regulatory Verification completed successfully.
 - Financial statements uploaded and validated.
 - AI Financial Analysis Engine available.
 - Business Rules Engine available.
 - Loan Origination System (LOS).
 - Document Management System (DMS).
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Assumptions

- Uploaded financial statements are authentic and readable.
 - AI models have been configured according to bank policy.
 - Threshold values for AI confidence are maintained by system administrators.
 - Financial ratio formulas are predefined by the bank.
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Out of Scope

- Manual editing of AI-generated financial calculations.
- External accounting software integration.
- Loan sanction decision.
- Credit Committee approval.
- Loan disbursement.
- Post-disbursement monitoring.